

[YIJC/2025/Q1]

- 1 More than 27000 spectators are expected to attend the National Day Parade (NDP). Each Singaporean can apply for 2, 4, or 6 tickets, with seats grouped together in the same row.

The seats are grouped into four coloured zones: Red (R), Green (G), Blue (B), and Yellow (Y). Each zone has 10 areas labelled from A to J. Each area has 20 rows (R01–R20) of 40 seats (01–40).

For example, a seat code 'B-E20-26' refers to a seat in the Blue Zone, Area E, Row 20 and Seat 26.

During the NDP, volunteers will be stationed at the entrances to assist in scanning the spectators' NRIC cards to retrieve their seat allocations and direct them to their seat locations.

Task 1.1

Write program code for a function `hash(string)` to compute and return the hash value for a given string, using the following algorithm:

- For each character in the string, calculate its ASCII value to the power of its position. The leftmost character in the string is at position 1
 - Sum all these values
 - Return the total modulo 1000.
- [2]

Task 1.2

Write program code to read the ticket information from `DATASET.TXT` and store all the information in a tuple of tuples named `data`. Each line in the text file contains an NRIC number, the number of tickets allocated, and the first seat code, separated by commas.

Each individual tuple should contain an NRIC number, the number of tickets allocated, and the first seat code. The output of `data` is similar to the following:

For example,

```
data = (('S9383870Z', 4, 'B-E20-36'),
        ('T1583391I', 2, 'B-C12-14'))
```

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Task 1.3

Write program code for a function `store(data)` to:

- create a hash table of size 1000
- for each individual tuple in `data`, compute the hash value for the NRIC number and use it as an index to locate the position in the hash table
- if the position is empty, store the individual data tuple at that position
- otherwise, at that position, create a list to store multiple individual tuples. [5]

To further enhance the security of personal data, a secured hash function `secured_hash(string)` is used to generate a secured hash table for the actual NDP event.

The following methods are provided in the Python file `secured_module.py`:

- `secured_hash(string)`: to securely hash the string and return an integer between 0 and 999 (inclusive)
- `data()`: to return the secured hash table of size 1000 containing all the ticket holders' information.

Task 1.4

Write program code for the function `query(tbl, nric)` to search the secured hash table `tbl` using the NRIC number and return the individual data in a tuple in the following format:

(NRIC number, number of tickets allocated, the first seat code)

If the NRIC number is not found in `tbl`:

- display the following statement:
`'No ticket was allocated to this NRIC number.'`
- return `None`.

Test Cases

```
>>> query(secured_hash_table, 'S9114928B')  
No ticket was allocated to this NRIC number.
```

```
>>> query(secured_hash_table, 'S9114928A')  
( 'S9114928A', 4, 'Y-I19-28' ) [5]
```

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Task 1.5

Write program code for a function `display(tup)` to display the information stored in the tuple `tup`.

Test Case

```
>>> tup = ('S9383870Z', 4, 'B-E20-36')
>>> display(tup)
```

Output:

```
'4 tickets allocated to NRIC: xxxxx870Z, and the seats are located
at the Blue Zone, Area E, Row 20, and Seats from 36 to 39.'
```

 [2]